To what extent do exports and gross capital formation account for cross-country variations in GDP, and which factor exhibits a larger elasticity relative to economic output?

Exploring the Relationship Between Exports, Investment and GDP

Introduction

From a macroeconomic perspective, international trade and capital accumulation are widely recognized as structural determinants of economic output. This project empirically investigates the relative contribution of exports and gross capital formation (investment) in explaining variations in cross-country GDP.

Research Question

To what extent do exports and gross capital formation account for cross-country variations in GDP, and which factor exhibits a larger elasticity relative to economic output?

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from sklearn.linear_model import LinearRegression
```

```
df = pd.read_csv("global_economy.csv")

df.head()
```

	CountryID	Country	Year	AMA exchange rate	IMF based exchange rate	Population	Currenc
0	4	Afghanistan	1970	0.044998	0.044998	10752971	Afgha
1	4	Afghanistan	1971	0.044998	0.044998	11015857	Afgha
2	4	Afghanistan	1972	0.044998	0.044998	11286753	Afgha
3	4	Afghanistan	1973	0.044998	0.044998	11575305	Afgha
4	4	Afghanistan	1974	0.044998	0.044998	11869879	Afgha

5 rows × 26 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10512 entries, 0 to 10511
Data columns (total 26 columns):
 #   Column
---  ------
 0    CountryID
 1    Country
 2    Year
 3    AMA exchange rate
 4    IMF based exchange rate
 5    Population
 6    Currency
 7    Per capita GNI
 8    Agriculture, hunting, forestry, fishing (ISIC A-B)
 9    Changes in inventories
 10   Construction (ISIC F)
 11   Exports of goods and services
 12   Final consumption expenditure
 13   General government final consumption expenditure
 14   Gross capital formation
 15   Gross fixed capital formation (including Acquisitions less dis
 16   Household consumption expenditure (including Non-profit instit
 17   Imports of goods and services
 18   Manufacturing (ISIC D)
 19   Mining, Manufacturing, Utilities (ISIC C-E)
 20   Other Activities (ISIC J-P)
 21   Total Value Added
 22   Transport, storage and communication (ISIC I)
 23   Wholesale, retail trade, restaurants and hotels (ISIC G-H)
 24   Gross National Income(GNI) in USD
 25   Gross Domestic Product (GDP)
dtypes: float64(20), int64(4), object(2)
memory usage: 2.1+ MB
```

```
df.columns
```

```
Index([' CountryID ', ' Country ', ' Year ', ' AMA exchange rate ',
       ' IMF based exchange rate ', ' Population ', ' Currency ',
       ' Per capita GNI ',
       ' Agriculture, hunting, forestry, fishing (ISIC A–B) ',
       ' Changes in inventories ', ' Construction (ISIC F) ',
       ' Exports of goods and services ', ' Final consumption
expenditure ',
       ' General government final consumption expenditure ',
       ' Gross capital formation ',
       ' Gross fixed capital formation (including Acquisitions less
disposals of valuables) ',
       ' Household consumption expenditure (including Non-profit
institutions serving households) ',
       ' Imports of goods and services ', ' Manufacturing (ISIC D)
',
       ' Mining, Manufacturing, Utilities (ISIC C–E) ',
       ' Other Activities (ISIC J–P) ', ' Total Value Added ',
       ' Transport, storage and communication (ISIC I) ',
       ' Wholesale, retail trade, restaurants and hotels (ISIC G–H)
',
       ' Gross National Income(GNI) in USD ',
       ' Gross Domestic Product (GDP) '],
      dtype='object')
```

```
df.describe()
```

	CountryID	Year	AMA exchange rate	IMF based exchange rate	Population
count	10512.000000	10512.000000	1.051200e+04	1.051200e+04	1.051200e+04
mean	431.116629	1996.262747	3.573959e+02	3.419846e+02	2.851523e+07
std	251.169090	14.900361	2.291128e+03	1.941857e+03	1.141296e+08
min	4.000000	1970.000000	4.300000e-14	4.300000e-14	4.359000e+03
25%	212.000000	1984.000000	1.000000e+00	1.000000e+00	6.330615e+05
50%	430.000000	1997.000000	2.812895e+00	2.761315e+00	5.051556e+06
75%	643.000000	2009.000000	5.134316e+01	4.806684e+01	1.678862e+07
max	894.000000	2021.000000	1.116366e+05	4.200000e+04	1.425893e+09

8 rows × 24 columns

Before conducting the analysis, I explored the dataset to understand its structure, variable types, and summary statistics.

```
df.isnull().sum()
```

	0
CountryID	0
Country	0
Year	0
AMA exchange rate	0
IMF based exchange rate	0
Population	0
Currency	0
Per capita GNI	0

Agriculture, hunting, forestry, fishing (ISIC A-B)	121
Changes in inventories	1841
Construction (ISIC F)	0
Exports of goods and services	21
Final consumption expenditure	0
General government final consumption expenditure	52
Gross capital formation	52
Gross fixed capital formation (including Acquisitions less disposals of valuables)	52
Household consumption expenditure (including Non-profit institutions serving households)	52
Imports of goods and services	42
Manufacturing (ISIC D)	43
Mining, Manufacturing, Utilities (ISIC C-E)	0
Other Activities (ISIC J-P)	0
Total Value Added	0
Transport, storage and communication (ISIC I)	49
Wholesale, retail trade, restaurants and hotels (ISIC G-H)	49
Gross National Income(GNI) in USD	0
Gross Domestic Product (GDP)	0

Missing values were checked to identify potential data quality issues before the analysis.

```
df.columns
```

```
Index([' CountryID ', ' Country ', ' Year ', ' AMA exchange rate ',
       ' IMF based exchange rate ', ' Population ', ' Currency ',
       ' Per capita GNI ',
       ' Agriculture, hunting, forestry, fishing (ISIC A–B) ',
       ' Changes in inventories ', ' Construction (ISIC F) ',
       ' Exports of goods and services ', ' Final consumption
expenditure ',
       ' General government final consumption expenditure ',
       ' Gross capital formation ',
       ' Gross fixed capital formation (including Acquisitions less
disposals of valuables) ',
       ' Household consumption expenditure (including Non-profit
institutions serving households) ',
       ' Imports of goods and services ', ' Manufacturing (ISIC D)
',
       ' Mining, Manufacturing, Utilities (ISIC C–E) ',
       ' Other Activities (ISIC J–P) ', ' Total Value Added ',
       ' Transport, storage and communication (ISIC I) ',
       ' Wholesale, retail trade, restaurants and hotels (ISIC G–H)
',
       ' Gross National Income(GNI) in USD ',
       ' Gross Domestic Product (GDP) '],
      dtype='object')
```

```
df.columns = df.columns.str.strip()

df.columns
```

```
Index(['CountryID', 'Country', 'Year', 'AMA exchange rate',
       'IMF based exchange rate', 'Population', 'Currency', 'Per
capita GNI',
       'Agriculture, hunting, forestry, fishing (ISIC A-B)',
       'Changes in inventories', 'Construction (ISIC F)',
       'Exports of goods and services', 'Final consumption
expenditure',
       'General government final consumption expenditure',
       'Gross capital formation',
       'Gross fixed capital formation (including Acquisitions less
disposals of valuables)',
       'Household consumption expenditure (including Non-profit
institutions serving households)',
       'Imports of goods and services', 'Manufacturing (ISIC D)',
       'Mining, Manufacturing, Utilities (ISIC C-E)',
       'Other Activities (ISIC J-P)', 'Total Value Added',
       'Transport, storage and communication (ISIC I)',
       'Wholesale, retail trade, restaurants and hotels (ISIC G-
H)',
       'Gross National Income(GNI) in USD', 'Gross Domestic Product
(GDP)'],
      dtype='object')
```

```
df[['Country',
    'Year',
    'Gross Domestic Product (GDP)',
    'Exports of goods and services',
    'Gross capital formation']].head()
```

	Country	Year	Gross Domestic Product (GDP)	Exports of goods and services	Gross capital formation
0	Afghanistan	1970	1.731436e+09	165618722.0	94611818.0
1	Afghanistan	1971	1.812838e+09	193580300.0	99012350.0
2	Afghanistan	1972	1.647900e+09	227654380.0	103456794.0
3	Afghanistan	1973	1.702716e+09	226913554.0	121728433.0

```
df[['Gross Domestic Product (GDP)',
    'Exports of goods and services',
    'Gross capital formation']].isnull().sum()
```

	0
Gross Domestic Product (GDP)	0
Exports of goods and services	21
Gross capital formation	52

dtype: int64

```
analysis_df = df[['Country', 'Year', 'Gross Domestic Product (GDP)'
                  'Exports of goods and services', 'Gross capital f

analysis_df = analysis_df[(analysis_df['Gross Domestic Product (GDP
                           (analysis_df['Exports of goods and servic
                           (analysis_df['Gross capital formation'] >
```

```
analysis_df['log_GDP'] = np.log(analysis_df['Gross Domestic Product
analysis_df['log_Exports'] = np.log(analysis_df['Exports of goods a
analysis_df['log_Investment'] = np.log(analysis_df['Gross capital f

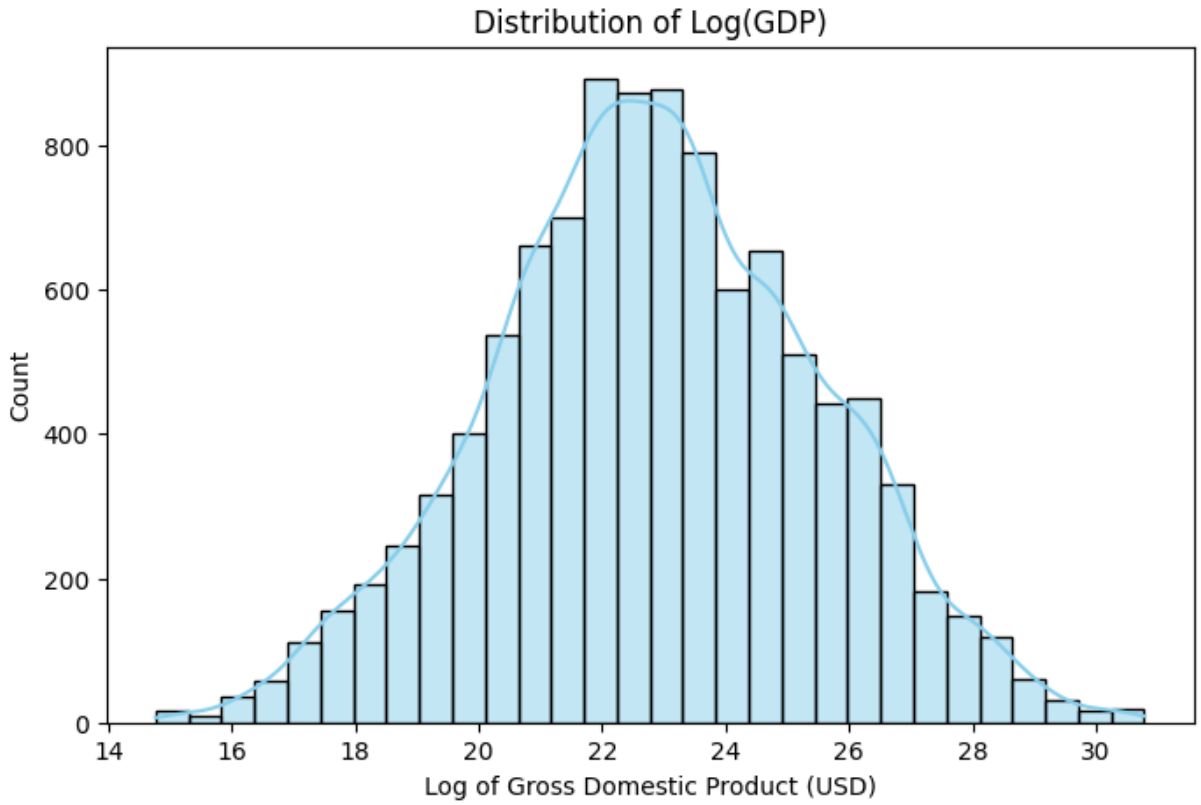
analysis_df[['log_GDP', 'log_Exports', 'log_Investment']].describe(
```

	log_GDP	log_Exports	log_Investment
count	10429.000000	10429.000000	10429.000000
mean	22.835983	21.617887	21.334803
std	2.646145	2.737332	2.706077
min	14.765303	12.134851	12.584389
25%	21.066638	19.812313	19.451489
50%	22.782239	21.620504	21.284608
75%	24.668069	23.470180	23.231746
max	30.779474	28.892319	29.659169

Economic theory suggests that exports and investment contribute to economic growth. This project investigates whether countries with higher export activity and investment levels also tend to have higher GDP.

The summary statistics provide an overview of the distribution and variation of GDP, exports, and investment levels across countries.

```
plt.figure(figsize=(8, 5))
sns.histplot(analysis_df['log_GDP'], bins=30, kde=True, color='skyb
plt.title('Distribution of Log(GDP)')
plt.xlabel('Log of Gross Domestic Product (USD)')
plt.ylabel('Count')
plt.show()
```



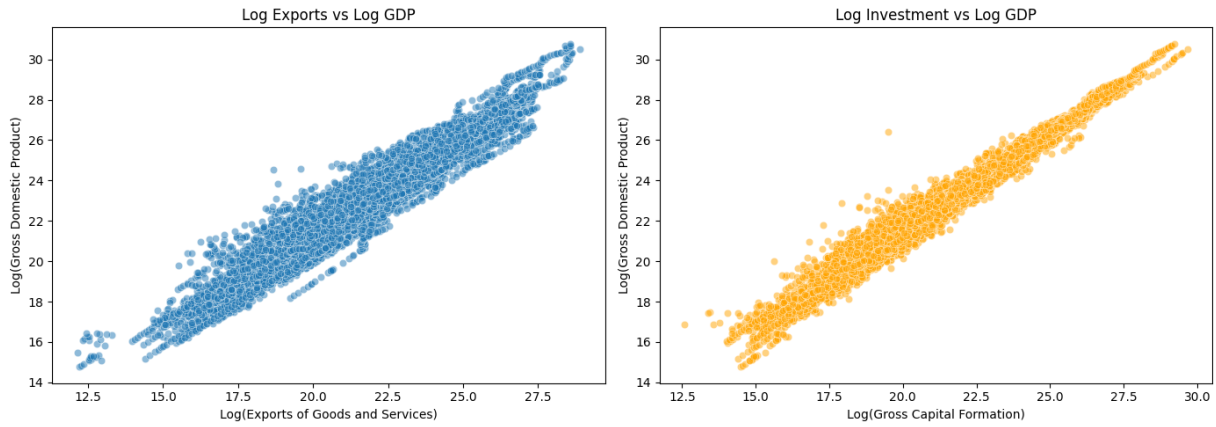
The distribution of log GDP appears relatively close to normal, although some skewness remains. The logarithmic transformation reduces the effect of extreme values and makes the distribution more suitable for statistical analysis.

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.scatterplot(data=analysis_df, x='log_Exports', y='log_GDP', alp
axes[0].set_title('Log Exports vs Log GDP')
axes[0].set_xlabel('Log(Exports of Goods and Services)')
axes[0].set_ylabel('Log(Gross Domestic Product)')


sns.scatterplot(data=analysis_df, x='log_Investment', y='log_GDP',
axes[1].set_title('Log Investment vs Log GDP')
axes[1].set_xlabel('Log(Gross Capital Formation)')
axes[1].set_ylabel('Log(Gross Domestic Product)')

plt.tight_layout()
plt.show()
```



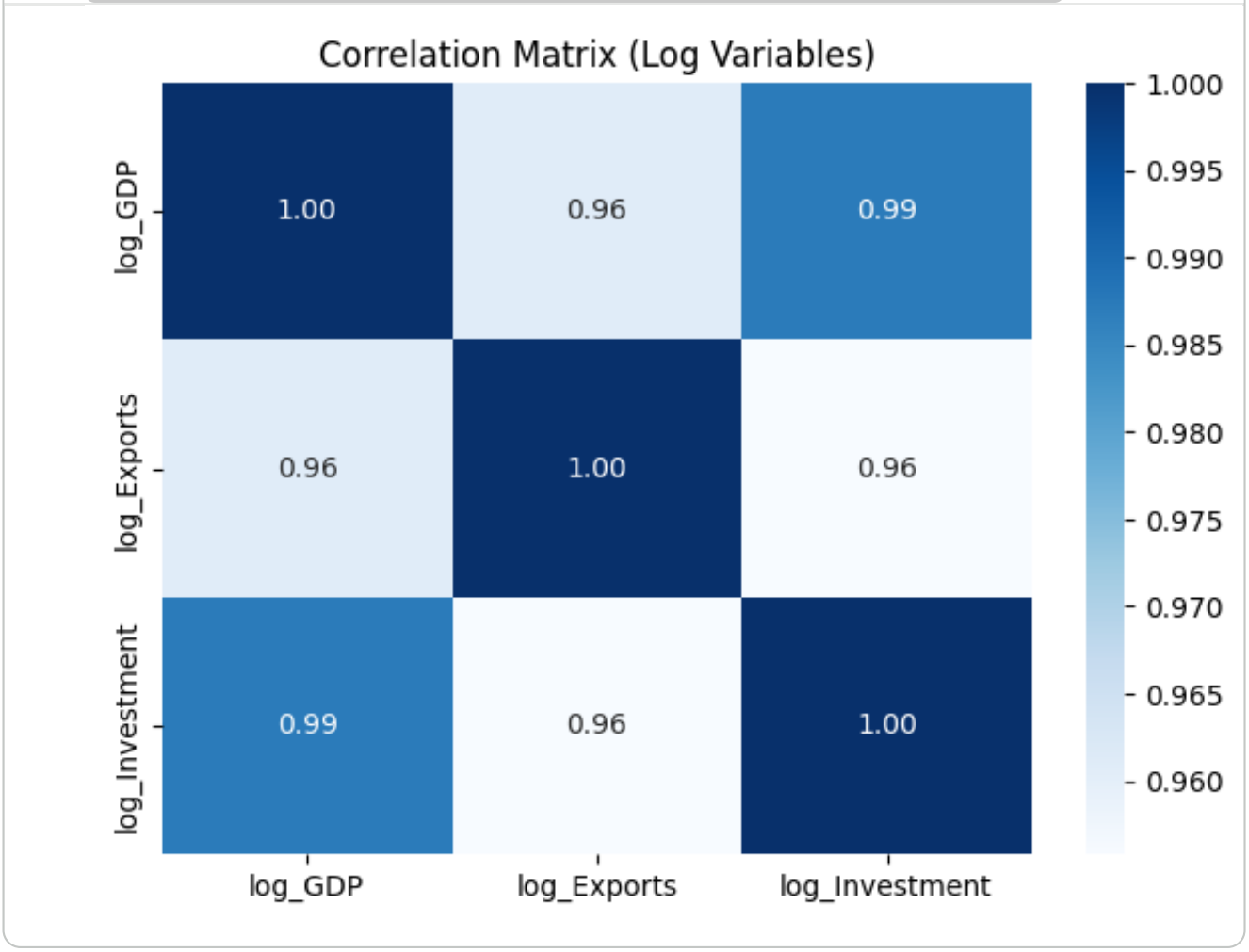
The scatter plot suggests a positive relationship between exports and GDP. Countries with larger export sectors generally tend to have larger economies. The relationship between investment and GDP appears slightly stronger and more linear than the relationship between exports and GDP.

```
analysis_df[
[
'Gross Domestic Product (GDP)',
'Exports of goods and services',
'Gross capital formation'
]
].corr()
#Before applying the log transformation, the baseline correlation c
```

	Gross Domestic Product (GDP)	Exports of goods and services	Gross capital formation
Gross Domestic Product (GDP)	1.000000	0.867134	0.945878
Exports of			

```
corr_matrix = analysis_df[['log_GDP', 'log_Exports', 'log_Investmen

plt.figure(figsize=(7, 5))
sns.heatmap(corr_matrix, annot=True, cmap='Blues', fmt=".2f")
plt.title('Correlation Matrix (Log Variables)')
plt.show()
```



The correlation matrix shows strong positive relationships between GDP, exports, and investment. In particular, GDP has a very strong correlation with gross capital formation and a strong correlation with exports.

```
X = analysis_df[['log_Exports', 'log_Investment']]
X = sm.add_constant(X) # add intercept term
y = analysis_df['log_GDP']

model = sm.OLS(y, X).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                log_GDP   R-squared:
Model:                            OLS   Adj. R-squared:
Method:                 Least Squares   F-statistic:
Date:                Wed, 07 Oct 2026   Prob (F-statistic):
Time:                        14:33:01   Log-Likelihood:
No. Observations:               10429   AIC:
Df Residuals:                   10426   BIC:
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                     coef    std err          t      P>|t|      [0.0
------------------------------------------------------------------------------
const              2.0482      0.031     66.477      0.000       1.9
log_Exports        0.1928      0.005     40.504      0.000       0.1
log_Investment     0.7790      0.005    161.767      0.000       0.7
==============================================================================
Omnibus:                     1536.454   Durbin-Watson:
Prob(Omnibus):                  0.000   Jarque-Bera (JB):
Skew:                           0.749   Prob(JB):
Kurtosis:                       6.056   Cond. No.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors
```

Regression Analysis Interpretation

The multiple linear regression results show that both log-transformed exports and log-transformed investment have statistically significant positive relationships with log GDP ($p < 0.001$).

Because log-log variables are used, the coefficients represent elasticity:

- A 1% increase in exports is associated with a percentage increase in GDP equal to the log_Exports coefficient, holding investment constant.
- A 1% increase in investment is associated with a percentage increase in GDP equal to the log_Investment coefficient, holding exports constant.

Together, these two variables account for a substantial proportion of the variance in global GDP (R-squared = 0.978).

log_Exports = 0.1928 log_Investment = 0.7790

The estimated coefficients suggest that a 1% increase in exports is associated with an approximate 0.19% increase in GDP, while a 1% increase in investment is associated with an approximate 0.78% increase in GDP, holding the other variable constant. This indicates that investment demonstrates a stronger relationship with GDP than exports in this dataset.

Discussion

The results are generally consistent with economic theory. Investment has a stronger relationship with GDP than exports in this dataset, as reflected by the larger coefficient. However, because the data includes different countries and years, the results should be interpreted as associations rather than evidence of causality.

Conclusion

This project investigated whether exports and investment are associated with GDP across countries.

The descriptive statistics, visualisations, correlation analysis and regression results suggest a positive relationship between exports, investment and economic output.

Countries with larger export sectors and higher levels of capital formation generally tend to exhibit higher GDP levels. However, these findings should be interpreted as associations rather than causal effects.

Additional variables and more advanced methods would be required to establish causality.

Overall, the findings suggest that exports and investment are important factors associated with differences in GDP across countries.

Limitations

This analysis has several limitations.

First, correlation does not imply causation.

Second, the dataset contains observations from different countries and years, which may have structural differences.

Third, other important factors affecting GDP, such as education, institutions and labour productivity, were not included.

Therefore, the results should be interpreted as descriptive evidence rather than proof of causal relationships.

I used AI only as a support tool during this project. Whenever I encountered coding errors or unexpected output, AI helped me understand the problem and suggested possible solutions. It was also used to check whether my code was running correctly. All final decisions and interpretations in this notebook are my own.